Scoring Combine-agent answers: RAGAS vs rubric vs key points

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RAGAS factual correctness disagrees with the rubric

`gmn-control-region-1`, mean of 3 runs per configuration

agent	FC recall	rubric (1–5)
GPT-5.6 luna, 3 configs	0.43–0.55	5.0
GPT-5.6 terra, 3 configs	0.41–0.66	5.0
Qwen 3.8, changed config	0.55	3.7

- 37 answers got rubric 5/5; their FC ranges 0.22 to 0.92
- reading them: the GPT answers are correct

What `FactualCorrectness(mode="recall")` computes
(ragas 0.2.15, `_factual_correctness.py`)

$$\text{recall} = \frac{TP}{TP + FN}$$

- TP: claims split from the **answer** that the gold supports
- FN: claims split from the **gold** missing in the answer
- two different LLM splits, redone on every call
- every gold claim weighs the same, side remarks included
  (e.g. the docs' "∼50%" comment in gmn)

Not "fraction of gold covered": it moves with answer length and claim split

Replacement: per-question key points ("nuggets")

Written once per question, frozen, versioned (`nuggets.yaml`):

`gmn-control-region-1`	
vital	use the `gmN` uncertainty type
vital	CR count N written after the keyword
vital	extrapolation factor α in the process column
vital	rate $= N\alpha$
okay	concrete datacard line; $1/\sqrt{N+1} \approx 45\%$
must_not	recommends `lnN`/`lnU`; example card inconsistent

Judge (Qwen 3.8 on aigw) per item:

- supported / partly / not supported
- must quote the answer; harness checks the quote is really there
- gold prose not shown; items shuffled; 3 judge runs, majority

Scores per answer:

$$V_{\text{strict}} = \frac{\#\,\text{vital fully supported}}{\#\,\text{vital}}$$

pass = all vital supported **and** no must_not present

Method of the TREC 2024/25 RAG track (AutoNuggetizer), plus evidence quotes and must_not items

Results: 2 questions × 8 agent configurations × 3 runs

question	agent / config	nugget pass	V_{strict}	FC recall	rubric
gmn	Qwen vanilla	0/3	0.00	0.00	1.0
	Qwen changed	2/3	0.67	0.55	3.7
	luna (3 configs)	3/3, 3/3, 3/3	1.00	0.43–0.55	5.0
	terra (3 configs)	2/3, 3/3, 3/3	1.00	0.41–0.66	5.0
shape-counting	Qwen vanilla	0/3	0.00	0.00	1.0
	Qwen changed	1/3	0.33	0.44	2.7
	luna van./v1/chg.	2/3, 1/3, 3/3	0.67, 0.50, 1.00	0.48, 0.60, 0.84	3.7, 3.7, 5.0
	terra van./v1/chg.	3/3, 3/3, 2/3	1.00, 1.00, 0.83	0.85, 0.80, 0.84	5.0

vanilla: shipped config (bash `ask`); v1: bash `deny` + read-only allow-list; changed: v1 + web fetch/search denied. Killed or empty runs count as 0.

- per answer: nugget pass agrees with "rubric ≥ 4" on 45/48 answers, with "FC ≥ 0.5" on 38/48
- correct GPT answers on gmn: $V_{\text{strict}} = 1.00$ where FC gave 0.41–0.55

Real answers, gmn: same rubric, different verdicts

GPT-5.6 luna, changed, run 2
FC 0.22 rubric 5 nugget pass (kp1–kp5 supported)

```
Use a gmN nuisance parameter. ...
rate          10.0    6.0
crStat   gmN  20   -    0.30
- 20 is the observed control-region count.
- 0.30 is the extrapolation factor.
- yield is therefore 20 x 0.30 = 6.0.
```

Correct and complete: FC calls it 22% of the gold.

GPT-5.6 terra, vanilla, run 2
FC 0.50 rubric 5 nugget fail (mn3 present)

```
Use a gamma-distributed (gmN) nuisance. ...
bkg_stat  gmN  N  -  alpha
For example, 25 CR events, alpha=0.12:
rate           3.0  3.0
bkg_stat       gmN  -    0.12
```

General rule right, but the example has no *N* after gmN: this card is wrong. Rubric 5, FC higher than the correct answer.

Excerpts of the final answers, unedited apart from "..." cuts and spacing.

Real answers, shape + counting card: diagnosis right vs wrong

GPT-5.6 luna, changed, run 1
FC 0.89 rubric 5 nugget pass (all 4 points)

```
‘counting‘ is a counting-only channel
and has no ‘shapes‘ mapping. ... fails:
KeyError: "Shape map has no entry for
           channel ’counting’"
... add  shapes * * FAKE  to the counting
card, or use  combineCards.py -S
I reproduced the failure ... and confirmed
that the -S version runs through FitDiagnostics.
```

Qwen 3.8, changed, run 1 (no tool calls)
FC 0.48 rubric 2 nugget fail (mn1 present, 0/2 vital)

```
These two datacard formats are
fundamentally incompatible: ...
combineCards.py can only combine datacards
of the same type ...
Convert the counting experiment to a shapes
datacard by creating a ROOT file ...
```

Wrong: `-S` exists. FC still gives it about half credit (0.48).

FC: correct answers 0.22–0.92, these two wrong ones 0.48 and 0.50. Nuggets separate them and name the reason.

What the nuggets add: a named reason for every failure

Caught, rubric said 5/5

- terra vanilla, gmn run 2: example card `bkg_stat gmN - 0.12` (no *N* after gmN) → must_not "inconsistent example" present

Diagnosed

- killed luna runs (permission reject, path typo): miss both vital points, not "low recall"
- Qwen changed, shape run 1: "shape and counting cards cannot be combined" → must_not

Missed / wrong (judge limits)

- terra changed, gmn run 2: `rate 10.0` with `gmN 25 0.12` ($25 \times 0.12 = 3.0$): arithmetic not caught
- 2 correct shape-counting answers failed: judge padded its quote with text not in the answer, quote check rejected it

rubric-vs-nugget disagreements	3/48
real error found by nuggets	1
judge false negative	2

Caveats and next steps

Caveats

- only 2 questions; key points written after seeing answers
- not yet calibrated against human labels
- Qwen judge grades Qwen answers (self-preference)
- 2 vital points on shape-counting: one judge flip $= \pm 0.5$
- agreement with humans is good for ranking systems, weak per question (TREC: τ 0.73–0.90 vs 0.3–0.66)

Next

- key points for all questions, drafted automatically from the **gold only** (LLM ensemble), then reviewed
- must_not items mined from observed wrong answers
- one-off calibration vs the 13 Oct. hand grades (Scott's π, judge leniency)
- cross-family judge for headline numbers
- report pass@1, pass^3, errors clustered by question

Proposal: nugget pass and V_{strict} as headline, rubric as second opinion, drop FC recall

References (scoring)

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